

Controller Rule Reference

Audit Reference	Rule Key	Type	Trigger / Condition	Controller Action
Phase initialization / reset	phase_initialization	initialization	New active phase uses that phase configured reward component initial weights.	No metric trigger. Initializes weights from PhaseConfig.reward_components.
Parseable Guard Fired	parseable_guard	guard update	parseable_answer_rate < 0.85.	Sets parseable_reward = max(current, 0.75).
Format Guard Fired	format_guard	guard update	solution_tag_compliance < 0.9 OR reasoning_tag_compliance < 0.88 OR malformed_answer_rate > 0.1.	Sets format_reward = max(current, 0.75).
Finish Guard Fired	finish_guard	guard update	truncation_rate > 0.15 OR average_completion_tokens > 0.85 * max_completion_length.	Adds +0.25 to finished_reward before bounds/floors are applied.
Stable Structure	stable_structure	helper condition	parseable >= 0.92 AND solution >= 0.95 AND reasoning >= 0.93 AND malformed <= 0.05 AND truncation <= 0.08.	Does not change weights by itself; required for correctness escalation.
Stable Window Ready	stable_window_ready	helper condition	At least 2 prior/current checkpoint metric entries exist in phase history.	Enables exact-match plateau check.
Correctness Plateau	correctness_plateau	helper condition	current normalized_exact_match minus previous window exact match < 0.02.	Does not change weights by itself; required for correctness escalation.
Correctness Rule Fired	correctness_escalation	update rule	stable_structure AND stable_window_ready AND correctness_plateau.	Adds +0.5 to correctness_reward before bounds/floors are applied.
Post-update Floors	component_floors	post-update constraint	Always applied after rule updates.	Ensures format_reward >= 0.25, parseable_reward >= 0.25, finished_reward >= 0.5.
Component Bounds Clamp	clamped_components	post-update constraint record	Any reward weight falls outside its RewardComponentConfig min/max bound.	Clamps to component min/max and records the changed component.

Notes. Helper conditions are recorded in controller decisions but do not directly update reward weights. Guard and escalation updates are followed by fixed component floors and component min/max clamping.